



Motivation

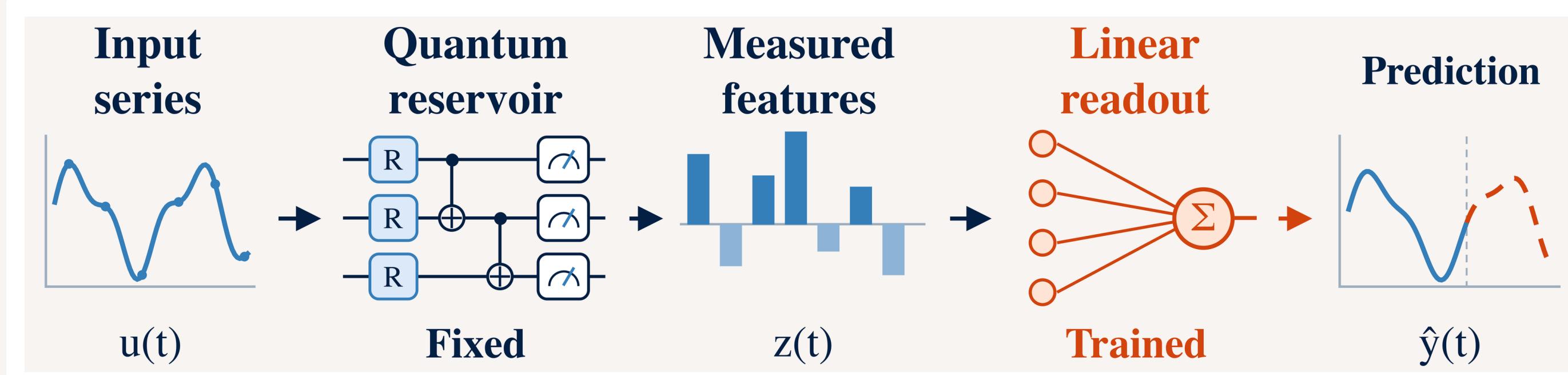
Can we make a QRC readout much smaller without discarding the structure of the measurements?

- ▶ A fixed quantum reservoir turns inputs into measurement features; **all learning happens in the classical readout**.
- ▶ Those features carry three natural axes: **time, qubits, and observables**.
- ▶ Flattening keeps the values but discards that organization; a dense neural readout then needs $H \, TnF$ weights.
- ▶ **ML-QRC learns one projection per mode**, then predicts from the compact tensor that results.

12.0× fewer parameters

33,513 vs. 403,329 parameters for the dense MLP readout, at equal hidden width $H = 64$.

Background

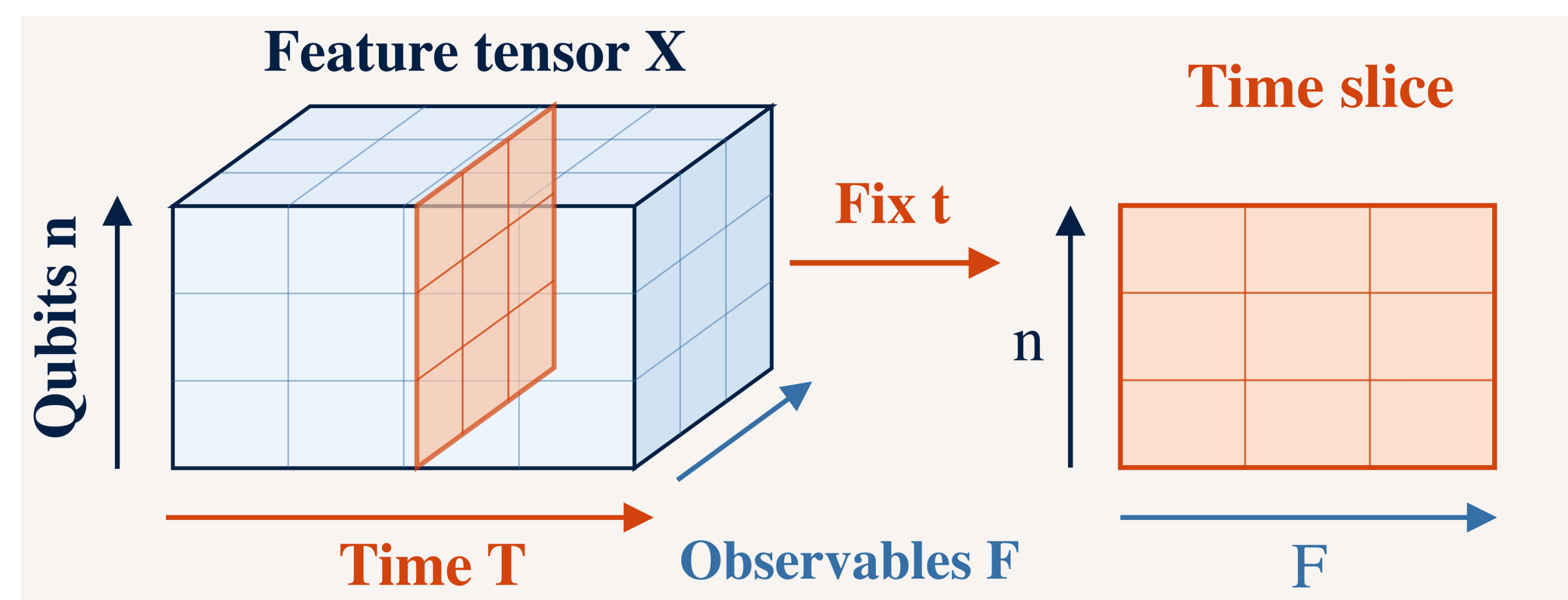


Quantum reservoir computing (QRC)

The input $u(t)$ is encoded into an n -qubit system evolving under a **fixed** reservoir unitary U_{res} [1, 2]. Because that unitary is never trained, QRC avoids the barren plateaus of variational circuits. Expectation values supply the classical features

$$\mathcal{X}[t, i, f] = \text{Tr}[\rho(t) O_f^{(i)}], \quad \mathcal{X} \in \mathbb{R}^{T \times n \times F}.$$

Time T Evolution across the input window
Qubits n Qubit-indexed measurements
Observables F Measured operator channels



Standard readouts

A flat Linear/Ridge readout acts directly on $\text{vec}(\mathcal{X})$. A dense neural readout instead learns

$$\mathbf{h} = \text{ReLU}(\mathbf{W}_h \text{vec}(\mathcal{X}) + \mathbf{b}_h), \quad \mathbf{W}_h \in \mathbb{R}^{H \times TnF}.$$

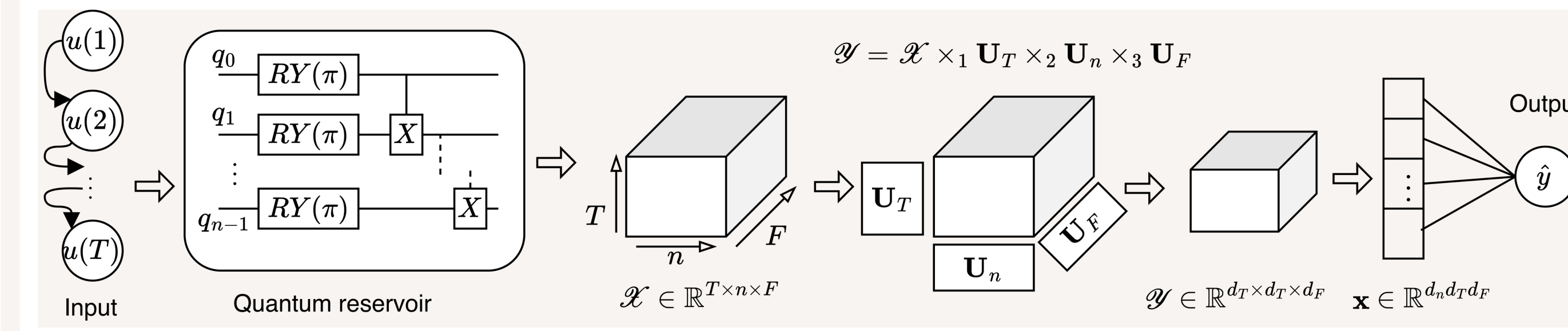
Its feature-to-hidden matrix holds $H \, TnF$ entries, **403,200** in the main configuration, and grows as $O(n^4)$ when $T = O(n)$ and $F = O(n^2)$. Flattening also treats every (t, i, f) triplet alike, despite the very different roles of temporal dynamics, qubit-qubit correlations, and observable basis.

Multilinear projections

Tensor contraction layers learn a separate map along each tensor mode [3]; quantum-based tensor contraction layers carry the idea into hybrid models, saving 98.68% of head parameters on VGG-19/CIFAR-100 [4].

Our setting: mode-wise projections learned on *quantum reservoir measurements*, with the reservoir held fixed.

Proposed method: ML-QRC



Fixed reservoir $\rightarrow$ measurement tensor $\rightarrow$ learned mode projections $\rightarrow$ neural prediction.

1. Keep the feature tensor intact

Collect the measurements as $\mathcal{X} \in \mathbb{R}^{T \times n \times F}$ rather than flattening them.

2. Learn one projection per mode

$$\mathcal{Y} = \mathcal{X} \times_1 \mathbf{U}_T \times_2 \mathbf{U}_n \times_3 \mathbf{U}_F, \\ \mathbf{U}_T \in \mathbb{R}^{d_T \times T}, \quad \mathbf{U}_n \in \mathbb{R}^{d_n \times n}, \quad \mathbf{U}_F \in \mathbb{R}^{d_F \times F},$$

so that every output entry mixes all three axes:

$$\mathcal{Y}[a, b, c] = \sum_{t, i, f} \mathcal{X}[t, i, f] U_T[a, t] U_n[b, i] U_F[c, f].$$

At $d_T = d_n = d_F = d = 8$ the representation goes from $50 \times 6 \times 21$ to $8 \times 8 \times 8$: the qubit mode is *widened* from 6 to 8, yet the whole tensor shrinks $6,300 \rightarrow 512$ entries ($12.3\times$).

3. Train projections and readout jointly

$$\mathbf{h} = \text{ReLU}(\mathbf{W}_h \text{vec}(\mathcal{Y}) + \mathbf{b}_h), \quad \hat{y} = \mathbf{W}_o \mathbf{h} + b_o,$$

with $H = 64$. Factors and head are optimized together by backpropagation; **the reservoir stays fixed**, so no variational quantum optimization is involved. The projection stage costs $O(d \, TnF)$, dominated by the first mode product.

Parameter count

For equal projection dimensions d and a single output,

$$P_{\text{MLP}} = H \, TnF + 2H + 1,$$

$$P_{\text{ML-QRC}} = d(T + n + F) + Hd^3 + 2H + 1.$$

Feature block: $403,200 \rightarrow 33,384$ ($12.1\times$).

Whole readout: $403,329 \rightarrow 33,513$ ($12.0\times$).

Compression improves as the reservoir grows

Qubits n	4	6	8	10
Flat dim. TnF	2,000	6,300	14,400	27,500
Dense MLP block	128,000	403,200	921,600	1,760,000
ML-QRC block	33,280	33,384	33,520	33,688
Compression	3.9×	12.1×	27.5×	52.2×

Feature-to-hidden block at $d = 8$, $T = 50$, $H = 64$, $F = n + \binom{n}{2}$. ML-QRC's cost is nearly independent of n because the Hd^3 head dominates.

Experimental setup

Mackey–Glass (MG). Chaotic forecasting, $\tau = 17$; 3,000 train / 1,000 test, plus a 10% regime (300 samples). Metric: NMSE $\downarrow$.

TFIM. Ferro- vs. paramagnetic ground-state classification, $g/J \in [0.2, 1.8]$ (8 + 8 couplings), 200 Z-basis sequences each; 3,200 samples, 70/30 split, plus a 50%-training regime. Metric: accuracy $\uparrow$.

Reservoir. Strongly entangling ansatz in PennyLane/PyTorch; $T = 50$, $n = 6$, $F = n + \binom{n}{2} = 21$ ($Z_i, Z_i Z_j$), so $TnF = 6,300$; analytic depolarizing channel, $p = 0.01$.

Baselines. Linear, Ridge, dense MLP, size-matched MLP-S (PCA + 64-unit MLP, 33,537 params), Ridge-PCA- d^3 (512 components), and Tensor-Ridge (fixed HOSVD ranks (8, 6, 8) + Ridge).

Training. Adam, lr 10^{-3} , weight decay 10^{-5} , 300 epochs, batch size 32; every neural head uses the same hidden width $H = 64$, so comparisons are made at matched head capacity. Results are mean $\pm$ std over **10 random seeds**.

Results and findings

Main comparison ($n = 6$, $d = 8$, 10 seeds)

Readout	Params	MG NMSE $\downarrow$		TFIM acc. % $\uparrow$	
		full	10%	full	50%
Linear	6,301	0.0218	0.6652	64.8	53.6 $\pm$ 2.1
Ridge	6,301	0.0085	0.7572	69.9	72.6 $\pm$ 1.0
MLP	403,329	0.0060 $\pm$ 0.0012	1.6105 $\pm$ 0.2529	69.5 $\pm$ 0.9	67.4 $\pm$ 1.4
MLP-S	33,537	0.0058 $\pm$ 0.0011	1.9463 $\pm$ 0.4839	70.4 $\pm$ 0.6	68.6 $\pm$ 1.5
Ridge-PCA- d^3	513	0.0085	0.7572	69.6	72.6 $\pm$ 0.5
Tensor-Ridge	385	0.0265	0.8258	67.0	62.1 $\pm$ 2.1
ML-QRC	33,513	0.0086 $\pm$ 0.0013	0.9684 $\pm$ 0.1581	70.5 $\pm$ 1.5	66.6 $\pm$ 1.8

Bold = best in column; closed-form readouts are seed-independent except on the TFIM split.

Classical context on MG: ESN [5] and NG-RC [6] are relevant classical reference architectures.

Projection dimension d ($n = 6$)

Config	d	Params	MG $\downarrow$	TFIM $\uparrow$
MLP (ref.)	—	403,329	0.0060	69.5
ML-QRC-S	4	4,533	0.0125	68.7
ML-QRC-M	8	33,513	0.0086	70.5
ML-QRC-L	16	263,505	0.0075	70.8
ML-QRC-XL	32	2,099,745	0.0067	71.4

Both metrics improve monotonically with d , but parameters grow as d^3 : S/M/L stay $89 \times / 12.0 \times / 1.5 \times$ below the dense MLP, XL is $5.2 \times$ larger.

What the results show

- ▶ **Full data.** On MG, ML-QRC (0.0086) has a similar mean NMSE to Ridge (0.0085); on TFIM it is numerically highest (70.5%), though inside the error bars of MLP-S and MLP.
- ▶ **Scarce data.** Among *neural* readouts ML-QRC is much the most stable on MG (0.97 vs. 1.61 and 1.95, both worse than predicting the training mean); Linear and Ridge still win outright, and Ridge takes TFIM at 50% data.
- ▶ **Learned factors beat fixed ones.** Against static-HOSVD Tensor-Ridge: MG $0.0265 \rightarrow 0.0086$, TFIM $67.0 \rightarrow 70.5\%$.
- ▶ **Qubit scaling.** Parameters stay nearly flat ($33,409 \rightarrow 33,817$ for $n = 4 \rightarrow 10$), so compression reaches $52\times$; but MLP-S accuracy scales better (78.4 vs. 74.1% on TFIM at $n = 10$).

Conclusions and open questions

Learned mode-wise projections deliver a 12.0× smaller neural readout that preserves the time–qubit–observable structure of QRC measurements, at competitive accuracy.

Limitations

- ▶ ML-QRC compresses *dense MLP* readouts; it does not replace Linear/Ridge, using $5.3\times$ more parameters.
- ▶ Its low-data stability edge holds only against neural readouts, and does not transfer to TFIM.
- ▶ Rank and size grow together in the d -sweep, so the two effects are confounded.

Next steps: finite-shot and hardware validation, larger reservoirs, lower-scaling tensor heads, and parameter-matched rank sweeps.

Selected references

- [1] K. Fujii, K. Nakajima, “Harnessing disordered-ensemble quantum dynamics for ML,” *Phys. Rev. Applied* 8, 024030, 2017.
- [2] K. Nakajima et al., “Boosting computational power through spatial multiplexing in QRC,” *Phys. Rev. Applied* 11, 034021, 2019.
- [3] J. Kossaifi et al., “Tensor contraction layers for parsimonious deep nets,” *CVPR Workshops*, 2017.
- [4] V. T. Nguyen, P. P. Markopoulos, “Quantum-based tensor contraction layers,” *Proc. SPIE* 14030, 1403001, 2026.
- [5] H. Jaeger, “The ‘echo state’ approach to analysing and training recurrent neural networks,” *GMD Report* 148, 2001.
- [6] D. J. Gauthier et al., “Next generation reservoir computing,” *Nature Comm.* 12, 5564, 2021.